

ORIGINAL RESEARCH

Multiclass Arrhythmia Detection and Classification From Photoplethysmography Signals Using a Deep Convolutional Neural Network

Zengding Liu , MPhil*; Bin Zhou, MD, PhD*; Zhiming Jiang, MPhil; Xi Chen, BE; Ye Li, PhD; Min Tang , MD, PhD†; Fen Miao , PhD†

BACKGROUND: Studies have reported the use of photoplethysmography signals to detect atrial fibrillation; however, the use of photoplethysmography signals in classifying multiclass arrhythmias has rarely been reported. Our study investigated the feasibility of using photoplethysmography signals and a deep convolutional neural network to classify multiclass arrhythmia types.

METHODS AND RESULTS: ECG and photoplethysmography signals were collected simultaneously from a group of patients who underwent radiofrequency ablation for arrhythmias. A deep convolutional neural network was developed to classify multiple rhythms based on 10-second photoplethysmography waveforms. Classification performance was evaluated by calculating the area under the microaverage receiver operating characteristic curve, overall accuracy, sensitivity, specificity, and positive and negative predictive values against annotations on the rhythm of arrhythmias provided by 2 cardiologists consulting the ECG results. A total of 228 patients were included; 118 217 pairs of 10-second photoplethysmography and ECG waveforms were used. When validated against an independent test data set (23 384 photoplethysmography waveforms from 45 patients), the DCNN achieved an overall accuracy of 85.0% for 6 rhythm types (sinus rhythm, premature ventricular contraction, premature atrial contraction, ventricular tachycardia, supraventricular tachycardia, and atrial fibrillation); the microaverage area under the microaverage receiver operating characteristic curve was 0.978; the average sensitivity, specificity, and positive and negative predictive values were 75.8%, 96.9%, 75.2%, and 97.0%, respectively.

CONCLUSIONS: This study demonstrated the feasibility of classifying multiclass arrhythmias from photoplethysmography signals using deep learning techniques. The approach is attractive for population-based screening and may hold promise for the long-term surveillance and management of arrhythmia.

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Key Words: arrhythmias ■ deep convolutional neural networks ■ photoplethysmography

Arrhythmias affect the quality of life of tens of millions of people worldwide, with as many as one-quarter of Americans over 40 years old developing cardiac arrhythmias.¹ Arrhythmias are associated with high risks of stroke,² heart failure,³ and even sudden cardiac death. More than 80% of sudden

Correspondence to: Fen Miao, PhD, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong 518055, People's Republic of China. Email: fen.miao@siat.ac.cn and Min Tang, MD, PhD, State Key Lab of Cardiovascular Disease, Fuwai Hospital, National Center for Cardiovascular Disease, Chinese Academy of Medical Sciences & Peking Union Medical College, No. 167 North Lishi Road, Xicheng District, Beijing 100037, People's Republic of China. Email: doctortangminfw@163.com

*Z. Liu and B. Zhou contributed equally.

†F. Miao and M. Tang contributed equally as co-senior authors.

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CLINICAL PERSPECTIVE

What Is New?

- The proposed end-to-end deep convolutional neural network model can triage photoplethysmography segments into sinus rhythm, premature ventricular contraction, premature atrial contraction, ventricular tachycardia, supraventricular tachycardia, and atrial fibrillation, with an overall accuracy of 85.0% (95% CI, 84.6%–85.4%).
- Our findings indicate that it is feasible to detect multiple arrhythmias from photoplethysmography signals with deep learning techniques.

What Are the Clinical Implications?

- A photoplethysmography-based approach to detect multiple arrhythmias is attractive for large-scale screening and has the potential to be helpful for the long-term and real-time monitoring of arrhythmia because of its convenience and real-time nature.
- Studies are needed to improve the accuracy of the proposed model and evaluate the performance of the model on data sets collected with wearable devices in nonhospital settings.

Nonstandard Abbreviations and Acronyms

DCNN	deep convolutional neural network
ML	machine learning
RFCA	radiofrequency catheter ablation
SR	sinus rhythm
SVT	supraventricular tachycardia

cardiac deaths are reportedly closely related to cardiac arrhythmias, and these deaths account for half of all deaths caused by heart diseases.⁴ However, some types of arrhythmias result in no symptoms, hindering timely detection in clinical settings.⁵

The conventional tools used to diagnose an arrhythmia include a 12-lead ECG and 24-hour Holter monitoring.^{6–8} However, ECG-based recorders are burdensome and inconvenient for patients because electrodes must be attached to the body. Unobtrusive and convenient methods for arrhythmia detection and monitoring would greatly promote the prevention and management of arrhythmia-related complications.⁹

Photoplethysmography is an emerging technology that offers a simple, inexpensive, and unobtrusive means to monitor the vascular system continuously.¹⁰ Several photoplethysmography-based methods have been used for arrhythmia detection and have yielded

promising results in previous studies.^{11–18} Such published photoplethysmography-based detectors have focused on the discrimination of atrial fibrillation (AF).¹⁹ By contrast, studies on detecting and classifying multiple arrhythmia types based on photoplethysmography signals have been few. Because different types of arrhythmias pose various risks to patients and require different clinical attention and intervention, accurately distinguishing the types would facilitate arrhythmia management.²⁰

To evaluate the accuracy of classifying multiple types of arrhythmias from photoplethysmography, we developed and validated a deep convolutional neural network (DCNN) for detecting and classifying multiple rhythm classes from raw photoplethysmography waveforms by using clinical annotations of rhythm based on ECG as the reference standard.

METHODS

Data Availability Statement

The validation set, test set, and code used in this study have been made publicly available at GitHub (<https://github.com/zdzdlu/PPGArrhythmiaDetection>). The training data are not available to other researchers.

Study Design

Between March 2020 and March 2021, patients with arrhythmia receiving radiofrequency catheter ablation (RFCA) at Fuwai Hospital, Chinese Academy of Medical Sciences, were recruited. The inclusion criteria for this study were (1) preoperative recording of arrhythmias, such as AF, premature atrial contraction (PAC), premature ventricular contraction (PVC), ventricular tachycardia (VT), and supraventricular tachycardia (SVT); (2) presence of indications for treatment with RFCA; (3) consent for RFCA; (4) patients without malignant tumors, severe organic heart diseases, or severe arterial stenosis; (5) intraoperative ECG and photoplethysmography signals collected using a multiparameter monitoring system. The exclusion criteria were (1) intraoperative detachment of an ECG electrode or photoplethysmography sensor and (2) recording time lasting <10 minutes. We have mastered the indications for RFCA based on the knowledge of guidelines and personal experiences.^{21–23} For AF (either paroxysmal AF or persistent AF), PAC, PVC, and VT, we performed RFCA in patients who were symptomatic when antiarrhythmic medications were ineffective, intolerant, or not desired by the patient. For SVT, we conducted RFCA in patients with symptomatic episodes. A multiparameter monitoring system (BeneVision N12; Shenzhen Mindray Bio-Medical Electronics) with simple 5-lead ECG wiring and a fingertip photoplethysmography sensor was used to

collect ECG and photoplethysmography data. Three lead ECG signals (I, II, and V) were used for the analysis, because they basically reflect the 3-dimensional vector of cardiac electrical conduction. The patient was in the supine position, and the sampling frequency used was 250 Hz. Informed consent was obtained from all patients who fulfilled the inclusion criteria. Clinical characteristics were abstracted from participants' electronic medical records. This study was approved by the Ethics Committee of Fuwai Hospital (approval number 2019-1239). The protocol was registered with the number ChiCTR2000031170 at www.chictr.org.cn.

Signal Preprocessing and Data Analysis

The collected signals were first preprocessed and then used for algorithm development, as illustrated in Figure 1. The raw ECG and photoplethysmography recordings were downsampled from 250 to 100 Hz and then bandpass filtered from 0.5 to 50 Hz and from 0.5 to 10 Hz, respectively. After being filtered, each ECG and photoplethysmography recording was split into 10-second-long nonoverlapping segments. The length of 10 seconds was chosen because it is large enough to identify a rhythm type and its promising results for classifying arrhythmias in the previous study.⁸ Subsequently, each 10-second photoplethysmography segment was normalized to ensure the segments all had the same scale. Normalized photoplethysmography segments corrupted by artifacts were removed using the signal quality evaluation algorithm proposed by Selvaraj et al.²⁴ Photoplethysmography segments with a noisy ECG reference were also excluded from the analysis

because no annotation could be made based on the reference data. Subsequently, all remaining 10-second ECG segments with sufficient signal quality were reviewed and annotated by 2 cardiologists blinded to the photoplethysmography results and each other's diagnosis based on the standardized ECG criteria.²⁵ Finally, all photoplethysmography segments were labeled following the diagnosis from the corresponding ECG.

Algorithm Development and Training

To detect multiple arrhythmia types from the photoplethysmography signals, we developed a DCNN model based on the popular convolutional neural network architecture VGGNet-16.²⁶ This architecture was selected because of its simple structure and powerful data representation ability. Because the traditional VGGNet-16 is designed for 2-dimensional image classification, we have modified it appropriately to accommodate the 1-dimensional input signal of this study. Details of the changes to the conventional VGGNet-16 are presented in Data S1. After adjustments, the architecture used for the DCNN model comprised 13 one-dimensional convolutional layers, 5 one-dimensional max-pooling layers, and 2 fully connected layers, as illustrated in Figure 2A. The specific structure, settings, and parameters of the DCNN model are summarized in Table S1. There are 1 496 102 trainable parameters in the model, requiring only about 5.71 MB of memory. The DCNN model accepted a 10-second photoplethysmography segment as its input and output a label prediction of one of the target classes.

The procedure for training, tuning, and evaluating the model is illustrated in Figure 2B. All patients' data were randomly

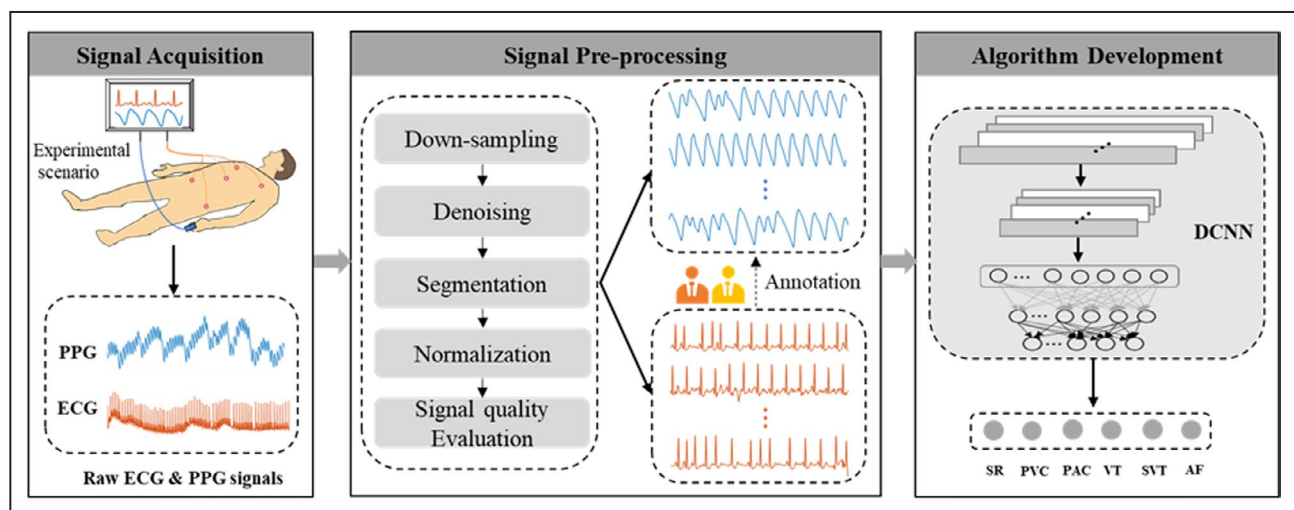


Figure 1. Overall structure of proposed method for detecting multiple arrhythmia types including signal acquisition, signal preprocessing, and algorithm development.

AF indicates atrial fibrillation; DCNN, deep convolutional neural network; PAC, premature atrial contraction; PPG, photoplethysmography; PVC, premature ventricular contraction; SR, sinus rhythm; SVT, supraventricular tachycardia; and VT, ventricular tachycardia.

partitioned into independent training, validation, and test data sets in the ratio 6:2:2. The training subset was used to train our DCNN model over 200 epochs. The model with optimal performance on the validation subset was saved and used for further evaluation with the test subset.

Statistical Analysis

Continuous variables with normal distribution were expressed as mean $\pm$ SD and compared using the *t* test. Nonnormally distributed continuous data were expressed as median (interquartile range) and compared using the Mann-Whitney test. Categorical variables were expressed as numbers (percentages) and compared using χ^2 or Fisher exact tests as appropriate. To verify the performance of the proposed DCNN model, traditional machine learning (ML)-based models, which rely on explicit rules and handcrafted features derived from interbeat intervals sequence and photoplethysmography waveforms, were used for comparison. Specifically, 17 handcrafted features and 4 ML algorithms (artificial neural network, random forest, k-nearest neighbors, and support vector machine) that have been used for photoplethysmography-based arrhythmia detection were used to develop the ML-based detectors, as shown in Figure S1. Each ML-based detector was trained and evaluated on the same training and test data sets as the DCNN model. A brief description of the ML-based detectors and the handcrafted features they rely on is available in Data S2.

For each arrhythmia detector, receiver operating characteristic curves were first generated for each rhythm to be detected, and a microaverage receiver operating characteristic curve was then created. The area under the microaverage receiver operating characteristic curve and its corresponding 95% CI were computed and compared for all arrhythmia detectors using the DeLong test.²⁷ Sensitivity, specificity, positive predictive value, negative predictive value (NPV), and overall accuracy were also calculated for each detector, along with the corresponding 95% CI.

All statistical tests in this study were 2-sided, and $P < 0.05$ was considered statistically significant. SPSS version 21.0 (IBM) and Python version 3.8.8 were used for data analysis.

Visualization of the Proposed DCNN

To improve the understanding of the DCNN's decisions, we used *t*-distributed stochastic neighbor embedding and guided gradient-weighted class activation mapping techniques for visual explanations. The *t*-distributed stochastic neighbor embedding technique is a nonlinear dimensionality reduction technique well suited for embedding high-dimensional data for visualization in a low-dimensional space.²⁸

The guided gradient-weighted class activation mapping combines fine-grained guided backpropagation and gradient-weighted class activation mapping. It produces a high-resolution class-discriminative heatmap from the final convolutional layer.²⁹ In this study, the features automatically learned from the various layers of the DCNN were mapped into 2-dimensional space using *t*-distributed stochastic neighbor embedding. The heatmap produced using guided gradient-weighted class activation mapping was superimposed over the photoplethysmography waveforms to highlight the regions in the photoplethysmography waveforms that were important to the DCNN for predicting a rhythm category.

RESULTS

Study Population

Two hundred forty-two patients with arrhythmia who met the inclusion criteria were enrolled in this study (Figure S2). Of the enrolled patients, 14 were excluded because of detachment of an ECG electrode ($N=5$) or photoplethysmography sensor ($N=3$) or because their recordings lasted <10 minutes ($N=6$). Two hundred twenty-eight pairs of photoplethysmography and ECG recordings were obtained from the consenting patients ($N=228$; 1 recording for each patient). The 228 photoplethysmography recordings were divided into 158 355 10-second photoplethysmography segments; 127 562 of these were retained, whereas another 30 793 (19.4%) were removed because of their poor signal quality or the poor signal quality of their ECG reference data. Among the 127 562 clean photoplethysmography segments, 118 217 (92.7%) were labeled as having a definite rhythm by the 2 cardiologists. The remaining segments were unclassified, because the 2 cardiologists could not provide the correct category.

Specifically, the segments numbered the following for each label: 38 081 for sinus rhythm (SR), 11 372 for PVC, 11 248 for PAC, 5783 for VT (3 or more consecutive PVCs at a rate of >100 beats per minute), 12 539 for SVT (3 or more consecutive PACs at a rate of >100 beats per minute), and 39 194 for AF. To train the model accurately, we removed unclassified segments. Consequently, 228 recordings with 118 217 clean 10-second photoplethysmography segments collected from 228 patients (age, 52.3 ± 11.3 years; 133 men) were retained in the final analysis. Each segment has only 1 identified rhythm type. Of the remaining 228 patients, we randomly separated 60% ($N=137$) into the training set, 20% ($N=46$) into the validation set, and 20% ($N=45$) into the test set. The segments included in the training, validation, and test sets were 71 390,

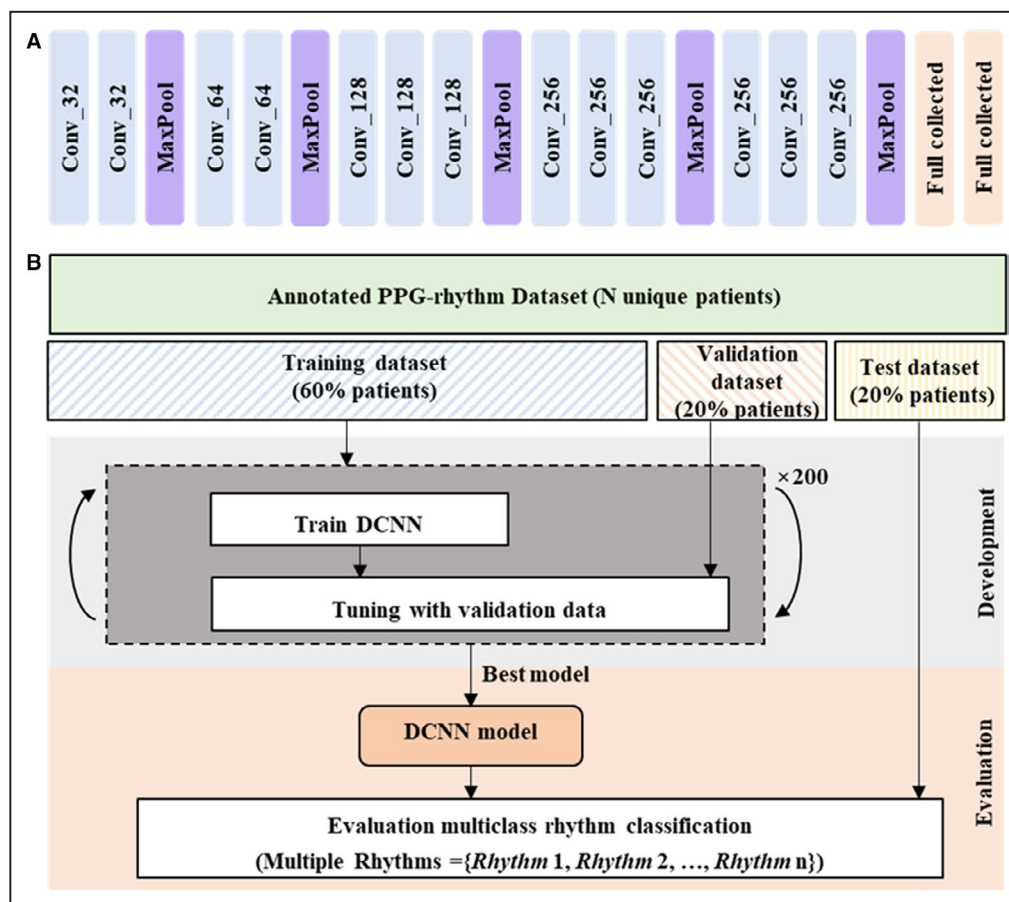


Figure 2. Architecture, development, and evaluation of the DCNN for classification of multiple arrhythmia types.

A, The DCNN model has 13 one-dimensional convolutional layers, 5 one-dimensional max-pooling layers, and 2 fully connected layers. **B**, Workflow illustrating the data sets used to train, tune, and evaluate the DCNN model. The symbol expressed as Conv_k represents a 1-dimensional convolution layer with k number of filters; for example, Conv_32 denotes a 1-dimensional convolution layer with 32 filters. DCNN indicates deep convolutional neural network; MaxPool, max-pooling; and PPG, photoplethysmography.

23 443, and 23 384, respectively. The baseline characteristics (Table 1) and the distribution of rhythm classes (Figure 3) were similar among the 3 data sets. Examples of 10-second synchronous I-lead ECG and photoplethysmography signal segments for various rhythm types are presented in Figure S3.

Multiclass Rhythm Classification

The performance of the DCNN in identifying multiple rhythms in the test set is summarized in Table 2; the corresponding confusion matrix and microaverage receiver operating characteristic results are provided in Figures 4A and 4B, respectively. The DCNN distinguished 6 rhythm types, SR, PVC, PAC, VT, SVT, and AF, in the photoplethysmography waveforms at an overall accuracy of 85.0% (95% CI, 84.6%–85.4%). Specifically, the average sensitivity, specificity, positive predictive value, and NPV for these 6 rhythm types were 75.8% (95% CI, 74.2%–77.4%), 96.9% (95% CI, 96.7%–97.1%), 75.2% (95%

CI, 73.7%–76.8%), and 97.0% (95% CI, 96.4%–97.6%), respectively.

According to the classification results of the 6 rhythms, the performances of the DCNN model in classifying 4 rhythms and 2 rhythms were further evaluated. For the classification of the 4 rhythms, PVC and PAC were considered as one category, and VT and SVT were considered as another category, whereas for the classification of the 2 rhythms, PVC, PAC, VT, SVT, and AF were considered as the same class (ie, non-SR). The overall accuracy and average sensitivity of the DCNN increased to 90.4% (95% CI, 90.1%–90.9%) and 88.2% (95% CI, 87.4%–89.1%) when classifying the 4 rhythms (Table S2) and to 97.8% (95% CI, 97.7%–98.0%) and 97.2% (95% CI, 96.9%–97.5%) when classifying the 2 rhythms (Table S3).

Table 3 shows a comparison of the performance of DCNN and conventional ML-based detectors in terms of overall accuracy and average sensitivity, specificity, positive predictive value, and NPV for the classification

Table 1. Baseline Characteristics of the Patients Included in the Study

	All patients, N=228	Training set, N=137	Validation set, N=46	P value*	Test set, N=45	P value†
Age, y	52.3±11.3	53.1±10.3	52.0±13.3	0.576	50.4±12.1	0.141
Men, n (%)	133 (58.3)	88 (62.2)	24 (52.2)	0.146	21 (46.7)	0.037
BMI, kg/m ²	25.5±2.9	25.5±2.4	25.3±3.2	0.701	25.5±4.0	0.958
SBP, mm Hg	130.4±14.8	130.3±12.6	132.8±21.4	0.344	127.9±12.8	0.273
DBP, mm Hg	82.5±10.2	82.9±9.3	82.9±14.4	0.967	80.8±7.4	0.187
CHA ₂ DS ₂ -VASc score	1.0 (0.0–2.0)	1.0 (0.0–2.0)	1.0 (1.0–3.0)	0.404	1.0 (0.3–2.0)	0.805
Hypertension, n (%)	38 (16.7)	20 (14.6)	12 (26.1)	0.076	6 (13.3)	0.833

Continuous variables are given as mean±SD or median (interquartile range). Categorical variables are given as n (percent). CHA₂DS₂-VASc score: heart failure=1 point; hypertension=1 point; age ≥75 years=2 points; diabetes=1 point; stroke=2 points; vascular disease=1 point; age between 65 and 74 years=1 point; gender (women)=1 point. BMI indicates body mass index; DBP, diastolic blood pressure; and SBP, systolic blood pressure.

*P value was obtained by comparison of the training and validation sets.

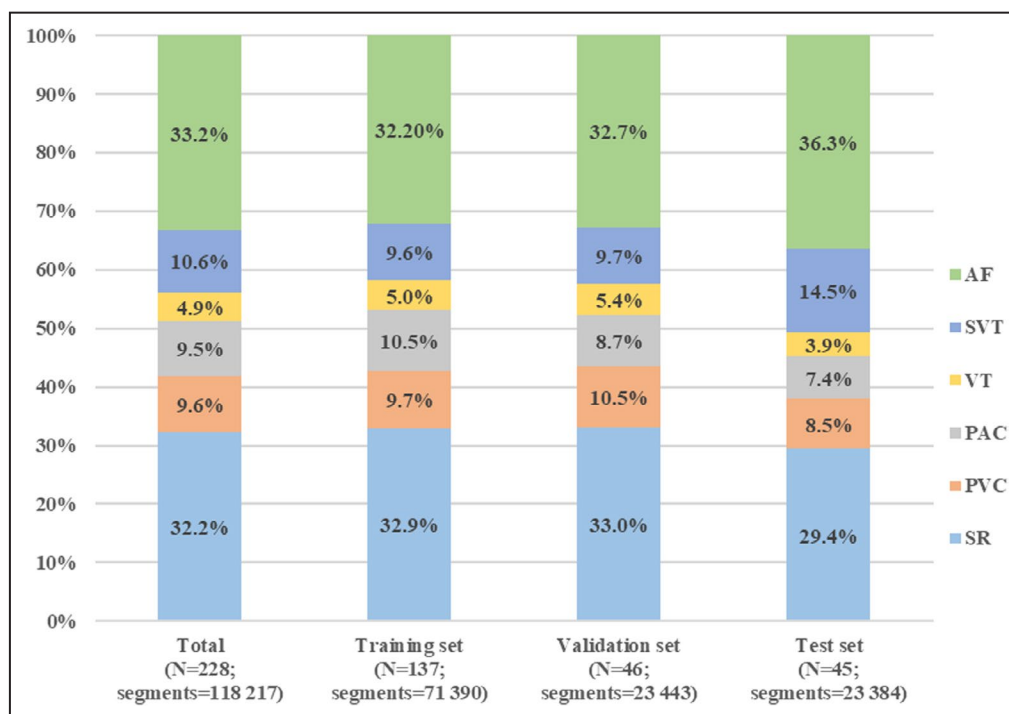
†P value was obtained by comparison of the training and test sets.

of 6 rhythms on the same test set. The detailed performance of the ML-based detectors for each rhythm type is summarized in Table S4. As indicated in Table 3, the ML-based detector obtained using the artificial neural network algorithm had the highest performance among the ML-based detectors in all evaluation metrics. Still, it did not perform as well as the DCNN did. The DCNN model outperformed the best ML-based model (the artificial neural network–based model) by 15.6%, 22.1%, 3.8%, 19.2%, and 3.3% in overall accuracy and average sensitivity, specificity, positive predictive value, and NPV, respectively. Moreover, the area under the microaverage receiver operating characteristic curve

of the DCNN was 0.978 (95% CI, 0.975–0.979), significantly higher than those of all ML-based detectors (area under the microaverage receiver operating characteristic curve, 0.891–0.922; $P<0.001$; Figure 4B).

Visualization of Learned Features

Figure 5 presents the *t*-distributed stochastic neighbor embedding visualizations of features learned from the various layers of the DCNN. In Figure 5, each point represents a photoplethysmography waveform mapped from the output features of a specific hidden layer in the DCNN into 2 dimensions. Points that belong to the same rhythm category are clustered. To balance the samples for a clearer

**Figure 3. Distribution of rhythm types across data sets.**

AF indicates atrial fibrillation; PAC, premature atrial contraction; PVC, premature ventricular contraction; SR, sinus rhythm; SVT, supraventricular tachycardia; and VT, ventricular tachycardia.

Table 2. Performance of the Deep Convolutional Neural Network for Classification of 6 Rhythms in the Test Set

	Value, % (95% CI)			
	Sensitivity	Specificity	PPV	NPV
SR	95.6 (95.2–96.2)	98.8 (98.6–98.9)	97.0 (96.6–97.4)	98.2 (97.9–98.5)
PVC	68.5 (66.4–70.5)	98.0 (97.9–98.2)	76.1 (74.3–78.2)	97.1 (96.5–97.8)
PAC	55.5 (53.2–57.9)	97.0 (96.8–97.2)	59.9 (57.5–62.3)	96.5 (95.6–97.3)
VT	72.7 (69.8–75.6)	96.4 (96.1–96.6)	44.9 (42.3–47.4)	98.9 (97.9–99.8)
SVT	68.0 (66.5–69.6)	97.8 (97.6–98.1)	84.1 (82.9–85.6)	94.2 (97.9–95.3)
AF	94.4 (93.9–94.9)	93.4 (93.0–93.8)	89.1 (88.4–89.7)	96.7 (96.2–97.2)
Average	75.8 (74.2–77.4)*	96.9 (96.7–97.1)*	75.2 (73.7–76.8)*	97.0 (96.4–97.6)*
Overall accuracy	85.0 (84.6–85.4)*			

AF indicates atrial fibrillation; NPV, negative predictive value; PAC, premature atrial contraction; PPV, positive predictive value; PVC, premature ventricular contraction; SR, sinus rhythm; SVT, supraventricular tachycardia; and VT, ventricular tachycardia.

*Significance level at 0.05.

display, we randomly selected 900 segments from each rhythm class (5400 in total) in the test set as a presentation data set and visualized the output features obtained at some representative layers (second, fourth, seventh, and 13th convolutional layers) of the DCNN. As illustrated in Figure 5, the degree of feature discrimination increased considerably overall with the depth of the DCNN.

Figure 6 illustrates examples of using the guided gradient-weighted class activation mapping approach to visualize crucial regions within the photoplethysmography signals that enabled the DCNN to predict a specific rhythm category. The I-lead ECG signals corresponding to the photoplethysmography are also shown. The darker color in Figure 6 indicates greater importance and a more significant contribution for the predicated category. The patterns specific to each rhythm have a larger attention weight. Specifically, the highlights of photoplethysmography in the cases of SR and AF were mainly focused on the positions of peaks that reflect the regularity of a pulse in waveforms. In the PVC and PAC cases, the waves with premature contractions appeared to be strongly highlighted. In the VT and SVT cases, the strongly highlighted waves were concentrated in consecutive PVC and PAC events that indicate the occurrence of VT and SVT.

DISCUSSION

This study validated the use of a deep learning-based approach to detect multiple rhythms from raw fingertip photoplethysmography waveforms obtained in a controlled hospital setting. We demonstrated that the DCNN performs well in diagnosing multiple types of rhythms based on photoplethysmography signals. Using ECG-based annotations as the reference, the DCNN achieved an overall accuracy of 85.0% in classifying 6 types of rhythms (5 arrhythmias included) from photoplethysmography signals and performed better than state-of-the-art nondeep

learning methods using handcrafted features. Our study indicated the feasibility of using a deep learning algorithm to detect multiple types of arrhythmias from photoplethysmography signals.

Previous Studies on Detection of Multiple Arrhythmia Types from Photoplethysmography Signals

Previous studies have used ML or deep learning techniques to detect arrhythmias from photoplethysmography signals. The ML-based solutions for arrhythmia detection from photoplethysmography signals involve calculating handcrafted features from a sequence of interbeat intervals and the photoplethysmography waveform. For instance, McManus et al proposed a linear detection algorithm that combined 2 statistical features calculated from the interbeat intervals sequences and achieved an accuracy of 96.8% in distinguishing AF versus SR from photoplethysmography signals.¹³ Sološenko et al investigated a feed-forward artificial neural network framework with 6 photoplethysmography features as inputs for discriminating PVC from SR in photoplethysmography signals and obtained a sensitivity of 98% and specificity of 98%.¹⁴ Polania et al used a support vector machine algorithm that combines frequency-domain and nonlinear dynamics features extracted from the interbeat intervals series and pulse amplitude information of photoplethysmography signals to distinguish between different arrhythmia types.¹⁵ Experimental results on 2 subjects showed that their proposed method identified PVC and VT from SR and PAC in photoplethysmography signals with 96.0% and 93.9% accuracy, respectively. These ML-based studies, however, have mainly focused on the discrimination between 2 rhythm types. Limited attention has been paid to the detection of multiple arrhythmia types.

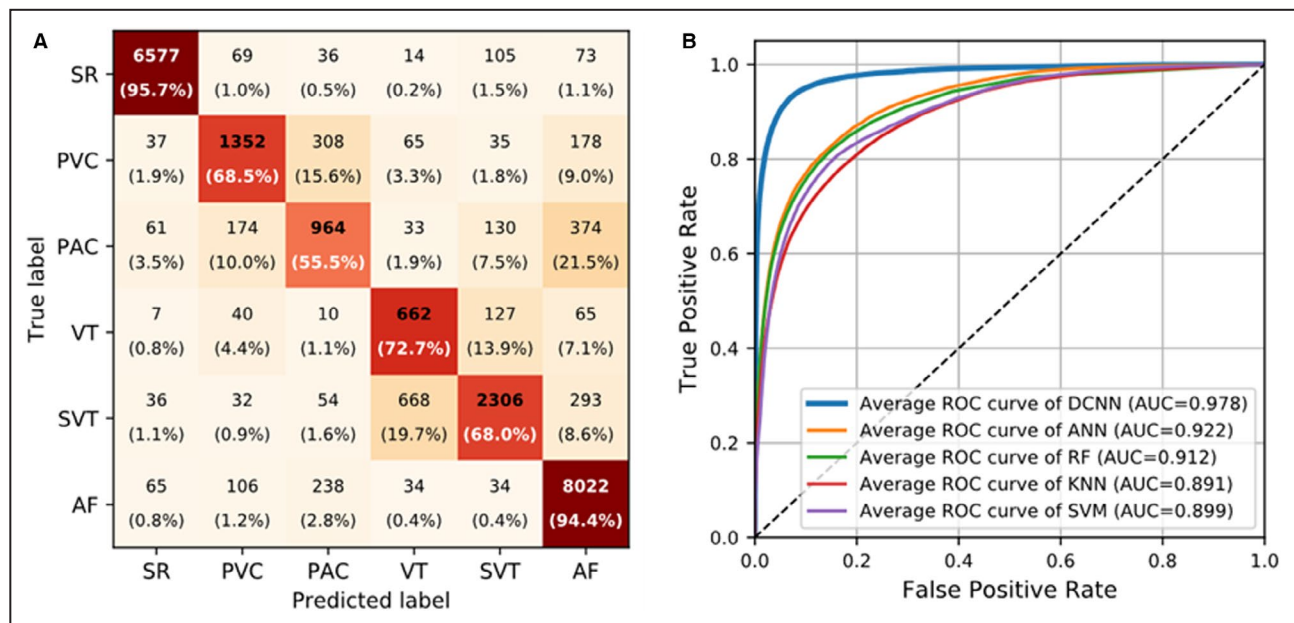


Figure 4. Results of the confusion matrix and ROC curves.

A, Confusion matrix with numbers and relative percentages to evaluate the performance of the DCNN for 6-rhythm discrimination. **B**, Microaverage ROC curves of the DCNN and ML-based detectors for 6-rhythm discrimination. In **(A)**, rows represent the categories given by the reference standard, and columns represent the categories predicted by the DCNN. Percentages were calculated by normalizing the results horizontally. AF indicates atrial fibrillation; ANN, artificial neural network; AUC, area under the ROC curve; DCNN, deep convolutional neural network; KNN, k-nearest neighbors; PAC, premature atrial contraction; PVC, premature ventricular contraction; RF, random forest; ROC, receiver operating characteristic; SR, sinus rhythm; SVM, support vector machine; SVT, supraventricular tachycardia; and VT, ventricular tachycardia.

In recent years, deep learning techniques, represented by DCNN, have achieved great success in computer vision.³⁰ These techniques simplify the feature extraction process (ie, automatically learn features from the original signal without the help of feature engineering) and provide an end-to-end solution for disease classification.¹⁹ With deep learning, solving the problem of detecting multiple types of arrhythmias from photoplethysmography signals is promising. Poh et al proposed a convolutional neural network architecture with 6 dense blocks to detect noise, SR, ectopy rhythm, and AF from photoplethysmography signals and reported an overall accuracy of 96.1%.¹⁶ Aliamiri et al developed a

hybrid convolutional neural network and recurrent neural network model for noise, AF, and non-AF segment classification on the basis of photoplethysmography signals and achieved a >99% area under the curve for AF detection.¹⁷ However, these studies did not validate the performance of their model in detecting more types of arrhythmias from photoplethysmography signals.

This study developed a deep learning model based on VGGNet-16 to detect multiple arrhythmia types from photoplethysmography signals. Our results revealed that our model significantly outperformed the ML-based models in detecting multiple rhythms from photoplethysmography signals. More importantly, we

Table 3. DCNN Performance for Classification of Multiple Rhythms Versus Several Machine Learning–Based Arrhythmia Classifiers

Method	Value, % (95% CI)				
	Average sensitivity	Average specificity	Average PPV	Average NPV	Overall accuracy
ML based					
ANN	53.7 (52.0–55.3)	93.1 (92.8–93.4)	56.0 (53.9–58.0)	93.7 (92.8–94.6)	69.4 (68.9–70.1)
RF	51.8 (50.1–53.5)	92.8 (92.4–93.1)	53.9 (52.0–55.9)	93.3 (92.4–94.2)	67.9 (67.3–68.5)
KNN	41.7 (40.2–43.1)	91.2 (90.9–91.6)	49.8 (47.1–52.4)	92.4 (91.4–93.3)	63.2 (62.6–63.8)
SVM	44.9 (43.4–46.3)	91.9 (91.5–92.2)	53.7 (50.7–56.8)	92.9 (92.0–93.8)	65.6 (65.0–66.2)
DCNN	75.8 (74.2–77.4)*	96.9 (96.7–97.1)*	75.2 (73.7–76.8)*	97.0 (96.4–97.6)*	85.0 (84.6–85.4)*

ANN indicates artificial neural network; DCNN, deep convolutional neural network; KNN, k-nearest neighbors; ML, machine learning; NPV, negative predictive value; PPV, positive predictive value; RF, random forest; and SVM, support vector machine.

*Significance level at 0.05.

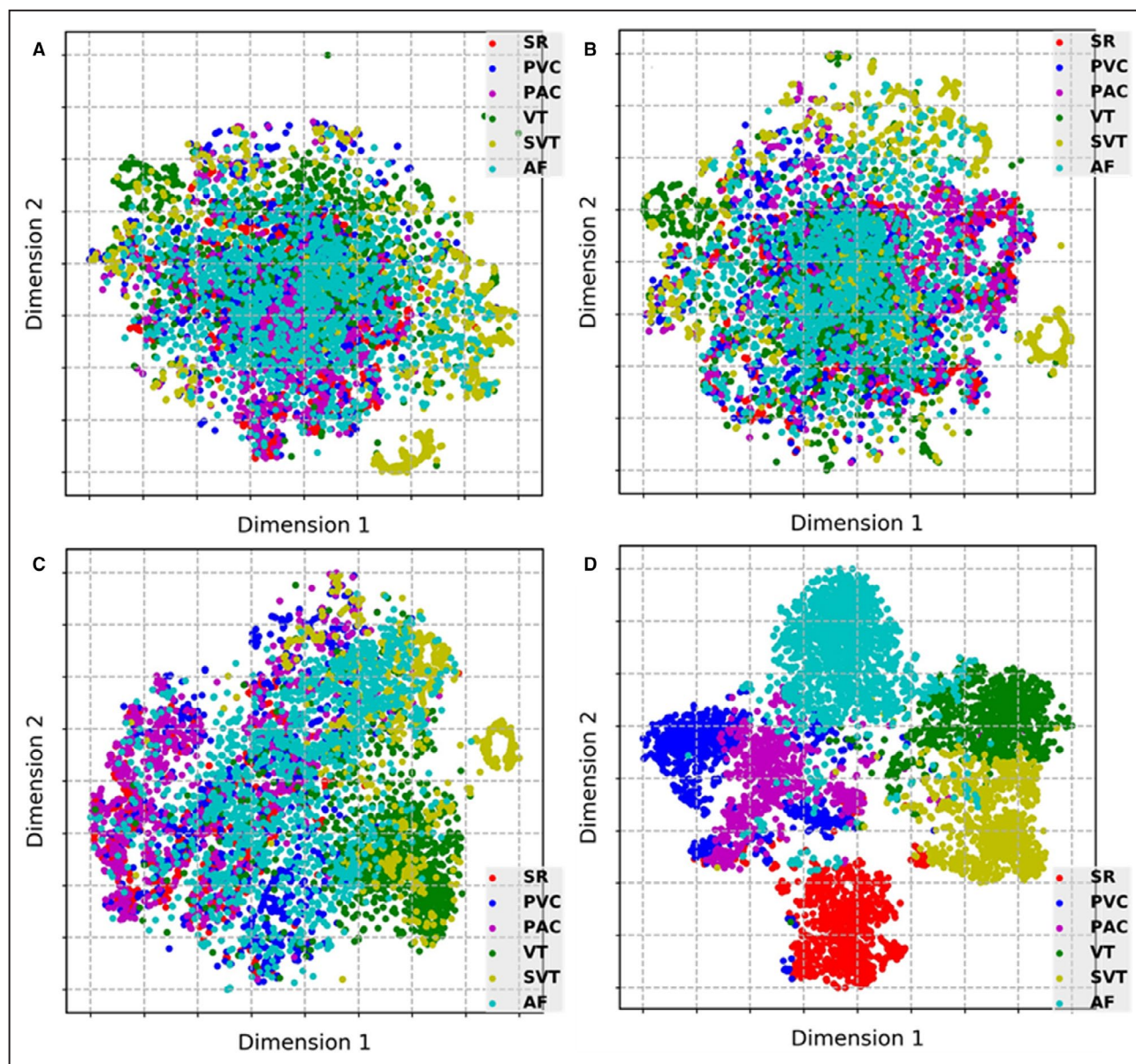


Figure 5. t-SNE visualizations of learned features from representative layers in the DCNN: (A) second, (B) fourth, (C) seventh, and (D) 13th convolutional layers

AF indicates atrial fibrillation; DCNN, deep convolutional neural network; PAC, premature atrial contraction; PVC, premature ventricular contraction; SR, sinus rhythm; SVT, supraventricular tachycardia; t-SNE, *t*-distributed stochastic neighbor embedding; and VT, ventricular tachycardia.

achieved an overall accuracy of 85.0% in classifying 5 types of arrhythmias (PVC, PAC, VT, SVT, and AF) and SR, thus validating the feasibility of detecting several arrhythmias through photoplethysmography signals.

Results of Multiple Arrhythmia Detection Based on Photoplethysmography in Our Study

PVC and PAC are similar premature heartbeats that disrupt the regular heart rhythm. Because of their

distinctive waveform morphologies in the ECG, PVC and PAC can be well detected by ECG data. However, because of the absence of ECG QRS- and P-wave information in the photoplethysmography, PVC and PAC (or VT and SVT) have similar patterns in the photoplethysmography waveforms, as shown in Figure 6. These similar patterns make it difficult to use photoplethysmography to distinguish PVC from PAC and thus VT from SVT. Therefore, PVC and PAC were usually regarded as 1 class to be detected in previous photoplethysmography-based studies to obtain

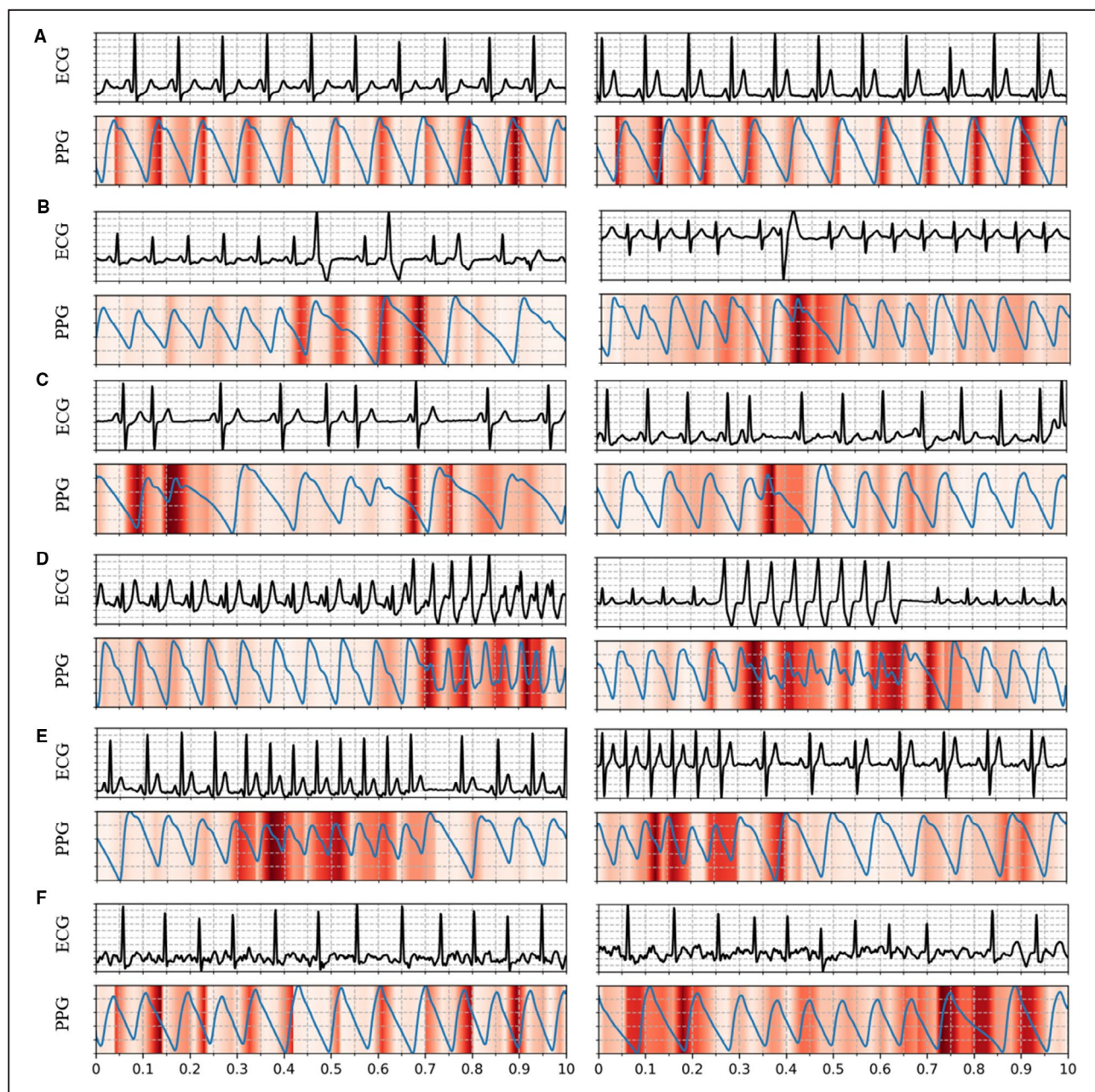


Figure 6. Examples of PPG waveforms with a Guided Grad-CAM visualization showing crucial regions for the DCNN to predict a certain triage category: (A) sinus rhythm, (B) premature ventricular contraction, (C) premature atrial contraction, (D) ventricular tachycardia, (E) supraventricular tachycardia, and (F) atrial fibrillation. In each panel, the I-lead ECG waveform corresponding to the PPG is also shown.

DCNN indicates deep convolutional neural network; Guided Grad-CAM: guided gradient-weighted class activation mapping; and PPG, photoplethysmography.

high accuracy.^{12,16} Our study found that the DCNN model showed a relatively low accuracy in detecting PVC, PAC, VT, and SVT compared with detecting SR and AF. Overall, 15.6% of PVCs and 21.5% of PACs were misidentified as PACs and AFs, respectively; 13.9% of VTs and 19.7% of SVTs were misclassified as SVTs and VTs, respectively; accounting for the first false-negative results of PVCs, PACs, VTs, and SVTs, as shown in Figure 4A. Therefore, the model must be

improved before clinical implementation. However, the present results are consistent with previous findings that ectopic rhythms are the main reason for false results when using photoplethysmography arrhythmia detection.^{16,18}

Notably, the main issue for a screening tool is to have a high sensitivity value. The average sensitivity of the DCNN model for detecting 6 rhythms was 75.8% (Table 2) and then improved to 88.2% (Table S3) and

97.2% (Table S4) for detecting 4 and 2 rhythms, respectively. However, the algorithm exhibited high specificity and NPV across all categories regardless of the number of rhythm types to be detected. High specificity and NPV, which can prevent causing needless angst in patients and reduce the costs of follow-up investigations, are also essential for a large-scale screening tool.³¹

As the most common cardiac arrhythmia, AF is one of the critical types identified by various arrhythmia detection methods. Compared with other studies that focused only on the determination of AF,^{12,13,17,18} more types of arrhythmias were included in our study. Moreover, even with a high burden of ectopic rhythms (ie, the segments of PVC, PAC, VT, and SVT [a group of rhythms that severely interfere with AF detection] accounted for 34.6% of the total segments, as shown in Figure 3), the DCNN revealed high accuracy (94.4% sensitivity and 93.4% specificity) in discriminating AF from photoplethysmography signals, implying the higher robustness of our model.

Clinical Application Prospect

The approach of this study has several potential clinical applications. First, the photoplethysmography-based approach offers a patient-friendly and convenient solution for arrhythmia detection. Although traditional ECG monitors are common and easily accessible in the clinic, the measurement of ECG signals requires electrodes to be attached to the body, resulting in discomfort and poor patient compliance, especially after wearing the device for a long time (eg, 24-hour Holter). Photoplethysmography measurement devices, however, can be worn more comfortably, because no additional accessories (eg, electrodes) are needed, making them more appealing and accessible to patients.

Second, photoplethysmography provides an unobtrusive method for continuous monitoring of arrhythmias in daily life. Although current ECG-based wearable devices can facilitate the measurement of ECG by reducing the number of electrodes, these devices still require active patient participation during implementation (eg, watch-based ECG monitors need users to place their fingers on the peripheral electrodes), resulting in the inability to monitor ECG continuously. Moreover, some patients with asymptomatic arrhythmias (eg, asymptomatic AF) may not take the initiative to measure their ECG, thus affecting the timely detection of arrhythmias. In contrast, photoplethysmography-based wearable devices enable long-term and real-time measurements of photoplethysmography without disturbing the patients and without requiring their compliance, thus ensuring continuous monitoring of the cardiac rhythm.

Third, photoplethysmography may be used for the prejudgment of arrhythmias before ECG makes

the final diagnosis. Today, several wearable devices have integrated both ECG and photoplethysmography sensors. During daily monitoring with these wearable devices, if the users actively measure the ECG after receiving an alert of an abnormal cardiac rhythm detected by the photoplethysmography, it will reduce the incidence of missed detection of irregular rhythms and allow for a precise diagnosis based on the ECG.

Limitations

This study has several limitations. First, to prevent any additional increase in time taken or cost incurred, we used a simple 5-lead ECG wire instead of a formal 12-lead ECG wire, and only 3-lead ECG signals (I, II, and V) were used for analysis. The reliance on 3-lead ECG signals to generate a reference diagnosis may produce false results. However, the 3-lead ECG signals (I, II, and V) reflected the 3-dimensional vector of cardiac electrical conduction and were reviewed by 2 cardiologists. Second, all photoplethysmography and ECG recordings were obtained using a commercial multiparameter monitor under controlled hospital conditions. The performance of the proposed DCNN has not been validated on photoplethysmography data sets collected from wearable devices used in everyday life. However, given the generalizability of the deep learning algorithm, the hardware used to acquire photoplethysmography signals is unlikely to affect the accuracy of the DCNN, provided the photoplethysmography recordings input to the model are of sufficient resolution and signal quality. Additionally, the DCNN only requires a photoplethysmography segment as short as 10 seconds, making it easy to obtain noise-free waveforms. Third, the primary arrhythmias included in this study were AF, SVT, VT, PAC, and PVC. Slow arrhythmias were not included in this study. Fourth, this was a single-center study with no additional independent test sets collected from other centers. Last, the model's accuracy for detecting PAC, PVC, SVT, and VT needs to be further improved. In the future, a multicenter study with a larger sample and arrhythmia type is required to develop a more accurate model. Furthermore, a test on photoplethysmography data sets collected using wearable devices in an unsupervised home setting is needed to reveal the performance of the proposed DCNN for multiclass arrhythmias detection.

CONCLUSIONS

This study evaluated the feasibility of using a deep learning algorithm to classify multiple arrhythmia types from photoplethysmography signals. Our results suggest that the proposed DCNN can, with high accuracy, classify photoplethysmography signals of 10-second duration into those indicating SR, PVC, PAC, VT, SVT,

and AF. The convenience of the photoplethysmography method supports a potential step toward developing a cost-effective tool for large-scale arrhythmia screening in nonhospital settings.

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Affiliations

Key Laboratory for Health Informatics, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China (Z.L., Z.J., X.C., Y.L., F.M.); University of Chinese Academy of Sciences, Beijing, China (Z.L.); Department of Cardiology, Laboratory of Heart Center, Zhujiang Hospital, Southern Medical University, Guangzhou, China (B.Z.); Fuwai Hospital, National Center for Cardiovascular Disease, State Key Lab of Cardiovascular Disease, National Clinical Research Center of Cardiovascular Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China (B.Z., M.T.); and Joint Engineering Research Center for Health Big Data Intelligent Analysis Technology, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China (Y.L.).

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Disclosures

None.

Supplemental Material

Tables S1-S4
Figures S1-S3
References 32–40

REFERENCES

- Lloyd-Jones DM, Wang TJ, Leip EP, Larson MG, Levy D, Vasan RS, D'Agostino RB, Massaro JM, Beiser A, Wolf PA, et al. Lifetime risk for development of atrial fibrillation: the framingham heart study. *Circulation*. 2004;110:1042–1046. doi: 10.1161/01.CIR.0000140263.20897.42
- Wolf PA, Dawber TR, Thomas HE, Kannel WB. Epidemiologic assessment of chronic atrial fibrillation and risk of stroke: the framingham study. *Neurology*. 1978;28:973. doi: 10.1212/WNL.28.10.973
- Wang TJ, Larson MG, Levy D, Vasan RS, Leip EP, Wolf PA, D'Agostino RB, Murabito JM, Kannel WB, Benjamin EJ. Temporal relations of atrial fibrillation and congestive heart failure and their joint influence on mortality: the framingham heart study. *Circulation*. 2003;107:2920–2925. doi: 10.1161/01.CIR.0000072767.89944.6E
- Mehra R. Global public health problem of sudden cardiac death. *J Electrocardiol*. 2007;40:S118–S122. doi: 10.1016/j.jelectrocard.2007.06.023
- Schmidt C, Kisselbach J, Schweizer PA, Katus HA, Thomas D. The pathology and treatment of cardiac arrhythmias: focus on atrial fibrillation. *Vasc Health Risk Manag*. 2011;7:193.
- Wang R, Fan J, Li Y. Deep multi-scale fusion neural network for multi-class arrhythmia detection. *IEEE J Biomed Health Inform*. 2020;24:2461–2472. doi: 10.1109/JBHI.2020.2981526
- Yao Q, Wang R, Fan X, Liu J, Li Y. Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network. *Information Fusion*. 2020;53:174–182. doi: 10.1016/j.inffus.2019.06.024
- Fan X, Yao Q, Cai Y, Miao F, Sun F, Li Y. Multiscaled fusion of deep convolutional neural networks for screening atrial fibrillation from single lead short ECG recordings. *IEEE J Biomed Health Inform*. 2018;22:1744–1753. doi: 10.1109/JBHI.2018.2858789
- Benjamin EJ, Chen P-S, Bild DE, Mascette AM, Albert CM, Alonso A, Calkins H, Connolly SJ, Curtis AB, Darbar D, et al. Prevention of atrial fibrillation: report from a national heart, lung, and blood institute workshop. *Circulation*. 2009;119:606–618. doi: 10.1161/CIRCULATIONAHA.108.825380
- Allen J. Photoplethysmography and its application in clinical physiological measurement. *Physiol Meas*. 2007;28:R1. doi: 10.1088/0967-3334/28/3/R01
- Chan PH, Wong CK, Poh YC, Pun L, Leung WWC, Wong YF, Wong MMY, Poh MZ, Chu DWS, Siu CW. Diagnostic performance of a smartphone-based photoplethysmographic application for atrial fibrillation screening in a primary care setting. *J Am Heart Assoc*. 2016;5:e003428. doi: 10.1161/JAHA.116.003428
- Bonomi AG, Schipper F, Erikäinen LM, Margarito J, van Dinther R, Muesch G, de Morree HM, Aarts RM, Babaeizadeh S, McManus DD, et al. Atrial fibrillation detection using a novel cardiac ambulatory monitor based on photo-plethysmography at the wrist. *J Am Heart Assoc*. 2018;7:e009351. doi: 10.1161/JAHA.118.009351
- McManus DD, Lee J, Maitas O, Esa N, Pidikiti R, Carlucci A, Harrington J, Mick E, Chon KH. A novel application for the detection of an irregular pulse using an iphone 4s in patients with atrial fibrillation. *Heart Rhythm*. 2013;10:315–319. doi: 10.1016/j.hrthm.2012.12.001
- Sološenko A, Petrénas A, Marozas V. Photoplethysmography-based method for automatic detection of premature ventricular contractions. *IEEE Trans Biomed Circuits Syst*. 2015;9:662–669. doi: 10.1109/TBCAS.2015.2477437
- Polanía LF, Mestha LK, Huang DT, Couderc J-P. Method for classifying cardiac arrhythmias using photoplethysmography. 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). 2015:6574–6577
- Poh M-Z, Poh YC, Chan P-H, Wong C-K, Pun L, Leung WW-C, Wong Y-F, Wong MM-Y, Chu DW-S, Siu C-W. Diagnostic assessment of a deep learning system for detecting atrial fibrillation in pulse waveforms. *Heart*. 2018;104:1921–1928. doi: 10.1136/heartjnl-2018-313147
- Aliamiri A, Shen Y. Deep learning based atrial fibrillation detection using wearable photoplethysmography sensor. *IEEE EMBS International Conference on Biomedical & Health Informatics (BHI)*. 2018;2018:442–445.
- Kwon S, Hong J, Choi E-K, Lee E, Hostallero DE, Kang WJ, Lee B, Jeong E-R, Koo B-K, Oh S, et al. Deep learning approaches to detect atrial fibrillation using photoplethysmographic signals: Algorithms development study. *JMIR mHealth and uHealth*. 2019;7:e12770. doi: 10.2196/12770
- Pereira T, Tran N, Gadhoumi K, Pelter MM, Do DH, Lee RJ, Colorado R, Meisel K, Hu X. Photoplethysmography based atrial fibrillation detection: a review. *NPJ Digital Medicine*. 2020;3:1–12. doi: 10.1038/s4174-019-0207-9
- Fu D-G. Cardiac arrhythmias: diagnosis, symptoms, and treatments. *Cell Biochem Biophys*. 2015;73:291–296. doi: 10.1007/s12013-015-0626-4
- Hindricks G, Potpara T, Dagres N, Arbelo E, Bax JJ, Blomström-Lundqvist C, Boriani G, Castella M, Dan GA, Dilaveris PE, et al. 2020 esc guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European association of cardiothoracic surgery (eacts). *Eur Heart J*. 2021;42:373–498. doi: 10.1093/eurheartj/ehaa612
- Al-Khatib SM, Stevenson WG, Ackerman MJ, Bryant WJ, Callans DJ, Curtis AB, Deal BJ, Dickfeld T, Field ME, Fonarow GC, et al. 2017 aha/acc/hrs guideline for management of patients with ventricular arrhythmias and the prevention of sudden cardiac death: Executive summary: a report of the American college of cardiology/American heart association task force on clinical practice guidelines and the heart rhythm society. *Circulation*. 2018;138:e210–e271. doi: 10.1161/CIR.0000000000000548
- Brugada J, Katritsis DG, Arbelo E, Arribas F, Bax JJ, Blomström-Lundqvist C, Calkins H, Corrado D, Devereux SG, Diller G-P, et al. 2019 esc guidelines for the management of patients with supraventricular tachycardia: the task force for the management of patients with supraventricular tachycardia of the European society of cardiology (esc). *Eur Heart J*. 2020;41:655–720. doi: 10.1093/eurheartj/ehz467
- Selvaraj N, Mendelson Y, Shelley KH, Silverman DG, Chon KH. Statistical approach for the detection of motion/noise artifacts in

- photoplethysmogram. *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. 2011;2011:4972–4975.
25. Issa ZF, Miller JM, Zipes DP. Clinical arrhythmology and electrophysiology: a companion to braunwald's heart disease. In: *Elsevier Health Sciences*. Philadelphia, PA: Saunders Elsevier; 2009.
 26. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. *arXiv*. Preprint posted online September 4, 2014. doi: 10.48550/arXiv.1409.1556
 27. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a non-parametric approach. *Biometrics*. 1988;837–845. doi: 10.2307/2531595
 28. Van der Maaten L, Hinton G. Visualizing data using t-sne. *J Mach Learn Res*. 2008;9:2579–2605.
 29. Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Grad-cam BD. Grad-cam: visual explanations from deep networks via gradient-based localization. In: *Proceedings of the IEEE international conference on computer vision*; 2017:618–626. doi: 10.1109/ICCV.2017.74
 30. Khan A, Sohail A, Zahoora U, Qureshi AS. A survey of the recent architectures of deep convolutional neural networks. *Artif Intell Rev*. 2020;53:5455–5516. doi: 10.1007/s10462-020-09825-6
 31. Altman DG. *Practical Statistics for Medical Research*. 1st ed. New York: Chapman & Hall/CRC; 1990. doi: 10.1201/9780429258589
 32. Eerikainen LM, Bonomi AG, Schipper F, Dekker LR, de Morree HM, Vullings R, Aarts RM. Detecting atrial fibrillation and atrial flutter in daily life using photoplethysmography data. *IEEE J Biomed Health Inform*. 2019;24:1610–1618. doi: 10.1109/JBHI.2019.2950574
 33. Tateno K, Glass L. Automatic detection of atrial fibrillation using the coefficient of variation and density histograms of rr and δrr intervals. *Med Biol Eng Compu*. 2001;39:664–671. doi: 10.1007/BF02345439
 34. Lake DE, Moorman JR. Accurate estimation of entropy in very short physiological time series: the problem of atrial fibrillation detection in implanted ventricular devices. *Am J of Physiol-Heart Circul Physiol*. 2011;300:H319–H325. doi: 10.1152/ajpheart.00561.2010
 35. Sarkar S, Ritscher D, Mehra R. A detector for a chronic implantable atrial tachyarrhythmia monitor. *IEEE Trans Biomed Eng*. 2008;55:1219–1224. doi: 10.1109/TBME.2007.903707
 36. Corino VD, Laureanti R, Ferranti L, Scarpini G, Lombardi F, Mainardi LT. Detection of atrial fibrillation episodes using a wristband device. *Physiol Meas*. 2017;38:787. doi: 10.1088/1361-6579/aa5dd7
 37. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Scikit-learn DV. Machine learning in python. *J Mach Learn Res*. 2011;12:2825–2830.
 38. Namdari A, Li Z. A review of entropy measures for uncertainty quantification of stochastic processes. *Adv Mech Eng*. 2019;11:1687814019857350. doi: 10.1177/1687814019857350
 39. Krivoshei L, Weber S, Burkard T, Maseli A, Brasier N, Kühne M, Conen D, Huebner T, Seeck A, Eckstein J. Smart detection of atrial fibrillation. *EP Europace*. 2017;19:753–757.
 40. Kugiumtzis D, Tsimpliris A. Measures of analysis of time series (mats): a matlab toolkit for computation of multiple measures on time series data bases. *arXiv*. Preprint posted online 9, Feb 2010. doi: 10.48550/arXiv.1002.1940

SUPPLEMENTAL MATERIAL

Data S1

Supplemental Methods

VGGNet-16 was designed for 2-dimensional image classification and had a kernel size of 3×3 . Some modifications to the traditional VGGNet-16 were required to accommodate the one-dimensional input signal in this study. First, the convolution and max-pooling layers were revised to be one-dimensional, with the kernel size being 1×3 . Second, the number of filters was reduced from 64, 128, 256, and 512 to 32, 64, 128, and 256, respectively. Third, the number of fully connected layers was reduced from 4 to 2. These steps were taken because when processing one-dimensional physiological signals, reducing the number of filters and fully connected layers can increase the training speed without affecting the performance of deep learning models. Finally, the deep convolutional neural network (DCNN) model employed in this study consists of 13 one-dimensional convolutional (Conv1d) layers, five one-dimensional max-pooling (MaxPool1d) layers, and two fully connected layers. Each convolutional layer was followed by a one-dimensional batch normalization (BatchNorm1d) layer and a rectified linear unit (ReLU) function. A ReLU layer and a dropout ($p = 0.5$) layer were applied between the fully connected layers. The detailed configuration of our DCNN model is provided in Table S1.

In the model, the weights were initialized by using the Kaiming initializer. The model was trained using Adam optimizer with default parameters and a mini-batch size of 128.

The learning rate was set to 0.001 at the beginning and then decayed exponentially ($\gamma = 0.95$) during the training. The construction and evaluation of the DCNN in this paper are implemented based on Pytorch in the CentOS7.3 operating system.

Data S2

The process of building the machine learning (ML)-based models for arrhythmia detection is shown in Figure S1. First, the position of peaks in the PPG waveform was detected (Figure S1A, left), and then the IPI series was obtained by calculating the time difference between two successive peaks (Supplementary Figure S1A, right). Second, the reported handcrafted features, including eight PPG waveform features (standard deviation value (STD), kurtosis, skewness, sample entropy (SampEn), Shannon entropy (ShEn), Hjorth mobility, Hjorth complexity, and spectral purity index (SPI)) and nine IPI features (mean value, STD, coefficient of variation (CoV), SampEn, ShEn, coefficient of sample entropy (COSEn), normalized root mean square of successive differences (nRMSSD) and point-care plot SD, and point-care plot SD2) were calculated from the PPG waveform and IPI series (Figure S1B).^{13, 32-35} Third, four ML algorithms that have been used for PPG-based arrhythmia detection were applied to construct the ML-based models (Figure S2C). The four ML algorithms include artificial neural network, random forest, k-nearest neighbors, and support vector machine.^{14, 15, 32, 36} All ML algorithms were implemented by using the Scikit-learn library in a Python programming environment.³⁷ The extracted features were described as follows:

■ Mean value and STD

Mean and standard deviation (STD) values are the most commonly used statistical parameters. We calculated the mean and STD of IPI sequences and the STD of the PPG waveform for arrhythmia detection.

■ CoV

CoV is a measure of relative variability.³³ It is the ratio of the standard deviation (σ) to the mean (μ), i.e., $CV = \frac{\sigma}{\mu}$.

■ Entropy measures

Entropy describes the confusion degree of a system. Two common indices, SampEn and ShEn are features of entropy theory. Previous studies have used the SampEn and ShEn to quantify the complexity of IPI sequences and PPG waveforms.^{13, 34} A detailed introduction of ShEn and SampEn can be found in.³⁸

■ COSEn

COSEn is an entropy estimate optimized for the detection of atrial fibrillation.³⁴ For a time series X , its COSEn can be calculated as

$$CosEn = SampEn - \ln(2r) - \ln(\mu) \quad (1)$$

Where $SampEn$ and μ are the SampEn and mean of X , respectively. r is the tolerance, usually set as $0.2 \cdot std(X)$.

■ nRMSSD

nRMSSD is the ratio of root mean square of successive difference to mean of IPI series,¹³ defined as

$$nRMSSD = \frac{1}{\sum_{i=1}^N a(i)} \cdot \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (a(i+1) - a(i))^2} \quad (2)$$

Where N is the length of IPI sequences, and $a(i)$ is the i th IPI, where $i = 1, 2, \dots, N$.

■ Poincaré plot parameters (SD1/SD2)

Poincaré plot, a scatter graph that shows the correlation between two consecutive data points in a time series, has been widely used for heart rate variability analysis. A Poincaré plot contains two important parameters, the length (SD1) and width (SD2) of the ellipse. Previously, Krivoshei *et al.* has studied SD1 and SD2 for discrimination of atrial fibrillation from PPG.³⁹ Given a time series with length N , $x_1, x_2, \dots, x_N$, the parameters SD1 and SD2 can be defined as:

$$SD1 = \frac{std(Y - X)}{\sqrt{2}} \quad (3)$$

$$SD_2 = \frac{std(Y + X)}{\sqrt{2}} \quad (4)$$

Where $X = (x_1, x_2, \dots, x_{N-1})$, $Y = (x_2, x_3, \dots, x_N)$, and $std(Y - X)$ and $std(Y + X)$ refer the standard deviation value of $Y - X$ and $Y + X$, respectively.

■ Kurtosis and Skewness

Kurtosis is a measure of the heavy or light tails of a normal distribution, and skewness measures the symmetry or asymmetry of data distribution. Kurtosis and skewness are defined as:

$$Kurtosis = E \left[\left(\frac{x - \mu}{\sigma} \right)^4 \right] \quad (5)$$

$$Skewness = E \left[\left(\frac{x - \mu}{\sigma} \right)^3 \right] \quad (6)$$

■ Hjorth parameters

Hjorth parameters called mobility ($\mathcal{H}_1$), complexity ($\mathcal{H}_2$), and spectral purity index (SPI) have been used for PPG-based atrial fibrillation detection.³² The parameters are calculated from the spectral moment of the signal. Let's define the n th order spectral moment as

$$m_n = \int_{-\pi}^{\pi} \omega^n S(\omega) d\omega \quad (7)$$

where $S(\omega)$ is the power spectrum. From the moment with different orders, $\mathcal{H}_1$, $\mathcal{H}_2$, and SPI are defined as the following formula:

$$\mathcal{H}_1 = \sqrt{\frac{m_2}{m_0}} \quad (8)$$

$$\mathcal{H}_2 = \sqrt{\frac{m_4}{m_2} - \frac{m_2}{m_0}} \quad (9)$$

$$SPI = \frac{m_2^2}{m_4 m_0} \quad (10)$$

Our study calculated the Hjorth parameters using a free MATLAB Toolkit provided by Kugiumtzis et al.⁴⁰

Table S1. Structure and configuration details of the DCNN.

Layers	Type	Output Size	Filters	Kernel-size	Strides	Padding	Number of Parameters
1	Conv1d	BS×32×1000	32	3	1	1	128
2	BatchNorm1d+ReLU	BS×32×1000	-	-	-	-	64
3	Conv1d	BS×32×1000	32	3	1	1	3 104
4	BatchNorm1d+ReLU	BS×32×1000	-	-	-	-	64
5	MaxPool1d	BS×32×333	-	3	3	0	0
6	Conv1d	BS×64×333	64	3	1	1	6 208
7	BatchNorm1d+ReLU	BS×64×333	-	-	-	-	128
8	Conv1d	BS×64×333	64	3	1	1	12 352
9	BatchNorm1d+ReLU	BS×64×333	-	-	-	-	128
10	MaxPool1d	BS×64×111	-	3	3	0	0
11	Conv1d	BS×128×111	128	3	1	1	24 704
12	BatchNorm1d+ReLU	BS×128×111	-	-	-	-	256
13	Conv1d	BS×128×111	128	3	1	1	49 280
14	BatchNorm1d+ReLU	BS×128×111	-	-	-	-	256
15	Conv1d	BS×128×111	128	3	1	1	49 280
16	BatchNorm1d+ReLU	BS×128×111	-	-	-	-	256
17	MaxPool1d	BS×128×37	-	3	3	0	0
18	Conv1d	BS×256×37	256	3	1	1	98 560
19	BatchNorm1d+ReLU	BS×256×37	-	-	-	-	512
20	Conv1d	BS×256×37	256	3	1	1	196 864
21	BatchNorm1d+ReLU	BS×256×37	-	-	-	-	512
22	Conv1d	BS×256×37	256	3	1	1	196 864
23	BatchNorm1d+ReLU	BS×256×37	-	-	-	-	512
24	MaxPool1d	BS×256×12	-	3	3	0	0
25	Conv1d	BS×256×12	256	3	1	1	196 864
26	BatchNorm1d+ReLU	BS×256×12	-	-	-	-	512
27	Conv1d	BS×256×12	256	3	1	1	196 864
28	BatchNorm1d+ReLU	BS×256×12	-	-	-	-	512
29	Conv1d	BS×256×12	256	3	1	1	196 864
30	BatchNorm1d+ReLU	BS×256×12	-	-	-	-	512
31	MaxPool1d	BS×256×4	-	3	3	0	0
32	Fully-connected	BS×256	-	-	-	-	262 400
33	Fully-connected	BS×6	-	-	-	-	1 542
Total parameters							1 496 102

Each conv1d layer is followed by a one-dimensional batch normalization (BatchNorm1d) layer and a rectified linear unit (ReLU) function. A ReLU layer and a dropout ($p = 0.5$) layer are applied between the fully connected layers. DCNN, deep convolutional neural network; Conv1d, one-dimensional convolutional; MaxPool1d, one-dimensional max-pooling; BS: batch size.

Table S2. Classification results of four rhythm types by the DCNN model on the test set.

	Value, % (95% CI)			
	Sensitivity	Specificity	PPV	NPV
SR	95.7 (95.2 to 96.2)	98.8 (98.6 to 98.9)	97.0 (96.6 to 97.4)	98.2 (97.9 to 98.5)
PVC and PAC	75.4 (74.0 to 76.8)	97.0 (96.8 to 97.3)	82.7 (81.4 to 84.0)	95.4 (94.9 to 96.0)
VT and SVT	87.5 (86.5 to 88.5)	97.6 (97.4 to 97.9)	89.3 (88.4 to 90.3)	97.2 (96.7 to 97.7)
AF	94.4 (93.9 to 94.9)	93.4 (93.0 to 93.8)	89.1 (88.4 to 89.7)	96.7 (96.2 to 97.2)
Average	88.2 (87.4 to 89.1)	96.7 (96.4 to 97.0)	89.5 (88.7 to 90.3)	96.9 (96.4 to 97.3)
Overall accuracy	90.4 (90.1 to 90.9)			

DCNN, deep convolutional neural network; SR, sinus rhythm; PVC, premature ventricular contraction; PAC, premature atrial contraction; VT, ventricular tachycardia; SVT, supraventricular tachycardia; AF, atrial fibrillation; PPV, positive predictive value; NPV, negative predictive value.

Table S3. Classification results of two rhythm types by the DCNN model on the test set.

	Value, % (95% CI)			
	Sensitivity	Specificity	PPV	NPV
SR	95.7 (95.2 to 96.2)	98.8 (98.6 to 98.9)	97.0 (96.6 to 97.4)	98.2 (97.9 to 98.5)
non-SR (PVC, PAC, VT, SVT, and AF)	98.8 (98.6 to 98.9)	95.7 (95.2 to 96.2)	98.2 (98.0 to 98.4)	97.0 (96.6 to 97.3)
Average	97.2 (96.9 to 97.5)	97.2 (96.9 to 97.5)	97.6 (97.3 to 97.9)	97.6 (97.3 to 97.9)
Overall accuracy	97.8 (97.7 to 98.0)			

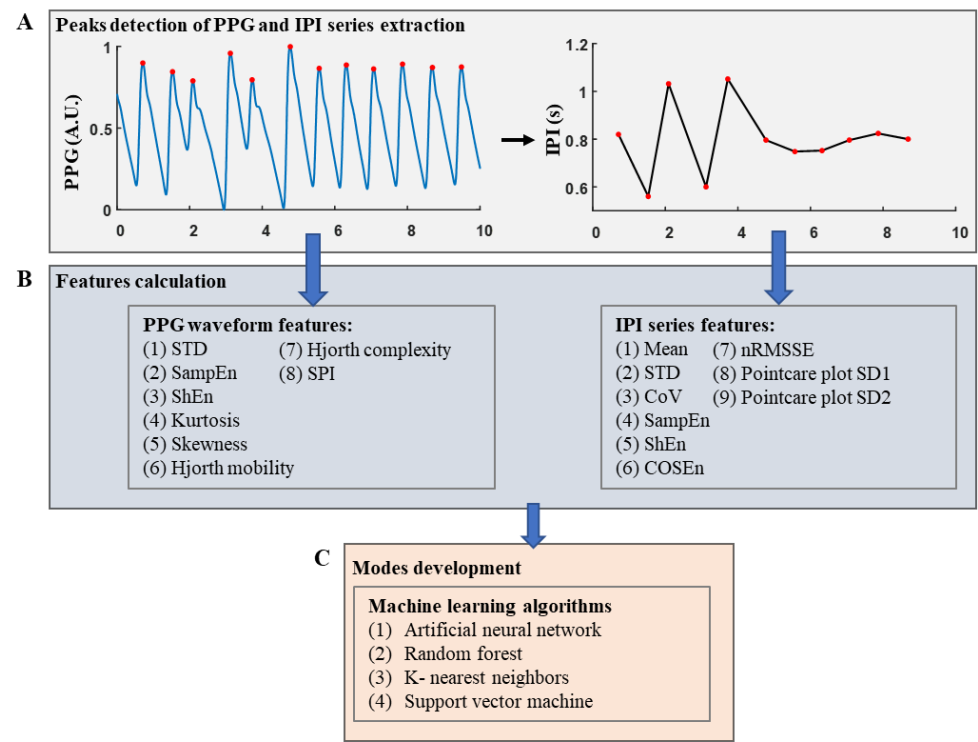
DCNN, deep convolutional neural network; SR, sinus rhythm; PVC, premature ventricular contraction; PAC, premature atrial contraction; VT, ventricular tachycardia; SVT, supraventricular tachycardia; AF, atrial fibrillation; PPV, positive predictive value; NPV, negative predictive value.

Table S4. Machine learning-based models for 6 rhythm types classification.

		Value, % (95% CI)			
		Sensitivity	Specificity	PPV	NPV
ANN					
	SR	82.1 (81.2 to 83.0)	93.5 (93.1 to 93.9)	84.0 (83.1 to 84.9)	92.6 (92.0 to 93.2)
	PVC	21.6 (19.8 to 23.4)	97.3 (97.1 to 97.5)	42.5 (39.4 to 45.5)	93.1 (92.1 to 94)
	PAC	14.6 (13.0 to 16.3)	97.6 (97.4 to 97.8)	32.5 (29.2 to 35.8)	93.4 (92.4 to 94.5)
	VT	58.4 (55.2 to 61.6)	96.0 (95.8 to 96.3)	37.4 (34.8 to 39.9)	98.3 (97.2 to 99.3)
	SVT	57.8 (56.2 to 59.5)	95.7 (95.5 to 96.0)	69.8 (68.1 to 71.4)	93.1 (92.3 to 93.8)
	AF	87.5 (86.8 to 88.2)	78.4 (77.7 to 79.1)	69.8 (68.9 to 70.7)	91.6 (90.7 to 92.6)
	Average	53.7 (52.0 to 55.3)	93.1 (92.8 to 93.4)	56.0 (53.9 to 58.0)	93.7 (92.8 to 94.6)
RF					
	SR	84.1 (83.2 to 84.9)	91.8 (91.4 to 92.2)	80.8 (79.9 to 81.8)	93.3 (92.7 to 94.0)
	PVC	19.7 (17.9 to 21.5)	97.0 (96.8 to 97.2)	37.9 (34.9 to 40.9)	92.9 (91.9 to 93.9)
	PAC	17.2 (15.4 to 19.0)	96.8 (96.5 to 97.0)	29.9 (27.1 to 32.8)	93.6 (92.5 to 94.6)
	VT	51.7 (48.5 to 54.9)	95.8 (95.6 to 96.1)	33.6 (31.1 to 36.0)	98.0 (96.9 to 99.1)
	SVT	54.7 (53.1 to 56.4)	96.4 (96.1 to 96.6)	71.9 (70.2 to 73.6)	92.6 (91.9 to 93.3)
	AF	83.5 (82.7 to 84.3)	78.9 (78.2 to 79.5)	69.4 (68.5 to 70.3)	89.3 (88.4 to 90.2)
	Average	51.8 (50.1 to 53.5)	92.8 (92.4 to 93.1)	53.9 (52.0 to 55.9)	93.3 (92.4 to 94.2)
KNN					
	SR	83.4 (82.5 to 84.3)	84.3 (83.7 to 84.9)	68.9 (67.9 to 69.9)	92.4 (91.6 to 93.3)
	PVC	8.4 (7.1 to 9.6)	98.9 (98.7 to 99.0)	40.3 (35.6 to 45.1)	92.1 (91.2 to 93.1)
	PAC	7.8 (6.6 to 9.1)	98.4 (98.2 to 98.5)	27.9 (23.9 to 31.9)	93.0 (92.0 to 94.1)
	VT	30.7 (27.7 to 33.7)	97.1 (96.9 to 97.3)	30.1 (27.1 to 33.0)	97.2 (96.1 to 98.3)
	SVT	33.7 (32.1 to 35.3)	97.3 (97.1 to 97.6)	68.2 (65.9 to 70.4)	89.7 (88.9 to 90.4)
	AF	86.1 (85.3 to 86.8)	71.5 (70.8 to 72.2)	63.3 (62.4 to 64.2)	90.0 (89.0 to 91.0)
	Average	41.7 (40.2 to 43.1)	91.2 (90.9 to 91.6)	49.8 (47.1 to 52.4)	92.4 (91.4 to 93.3)
SVM					
	SR	85.3 (84.5 to 86.2)	86.1 (85.6 to 86.7)	71.9 (71.0 to 72.9)	93.4 (92.6 to 94.2)
	PVC	7.3 (6.1 to 8.4)	99.5 (99.4 to 99.6)	59.3 (53.1 to 65.4)	92.1 (91.3 to 92.9)
	PAC	4.2 (3.3 to 5.1)	99.2 (99.0 to 99.3)	28.5 (23.0 to 34.0)	92.8 (91.8 to 93.9)
	VT	39.4 (36.2 to 42.6)	96.7 (96.4 to 96.9)	32.3 (29.6 to 35.1)	97.5 (96.5 to 98.6)
	SVT	47.0 (45.3 to 48.7)	95.7 (95.5 to 96.0)	65.1 (63.2 to 67.0)	91.4 (90.7 to 92.2)
	AF	85.9 (85.2 to 86.6)	74.0 (73.3 to 74.7)	65.3 (64.4 to 66.2)	90.2 (89.2 to 91.2)
	Average	44.9 (43.4 to 46.3)	91.9 (91.5 to 92.2)	53.7 (50.7 to 56.8)	92.9 (92.0 to 93.8)

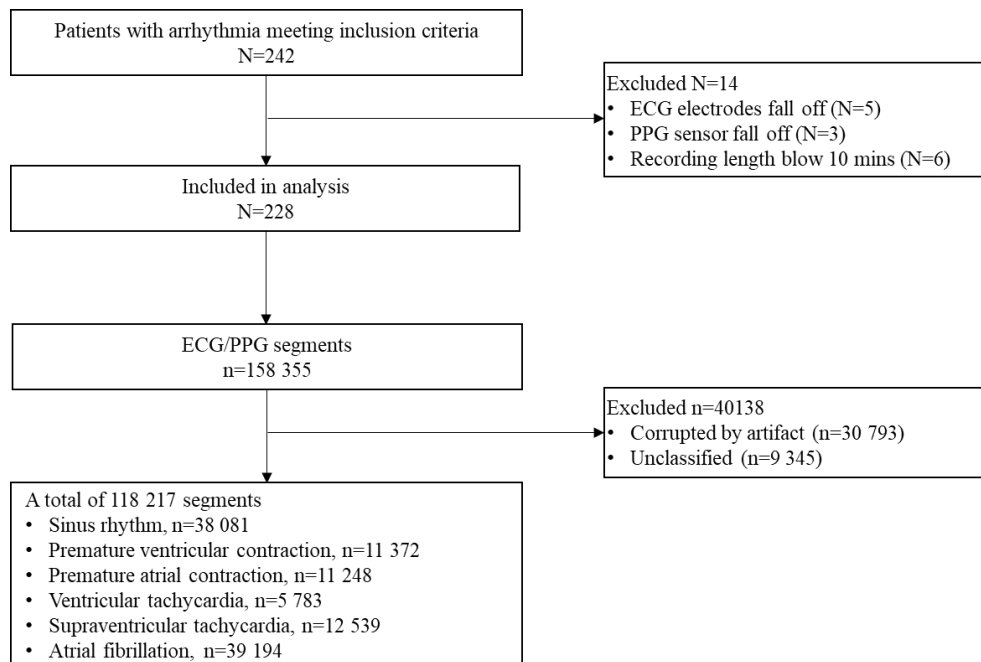
SR, sinus rhythm; PVC, premature ventricular contraction, PAC, premature atrial contraction, VT, ventricular tachycardia; SVT, supraventricular tachycardia; AF, atrial fibrillation; ANN, artificial neural network; RF, random forest; KNN, k- nearest neighbors; SVM, support vector machine; PPV, positive predictive value; NPV, negative predictive value.

Figure S1. Processes for machine learning-based arrhythmia detection models’ development.



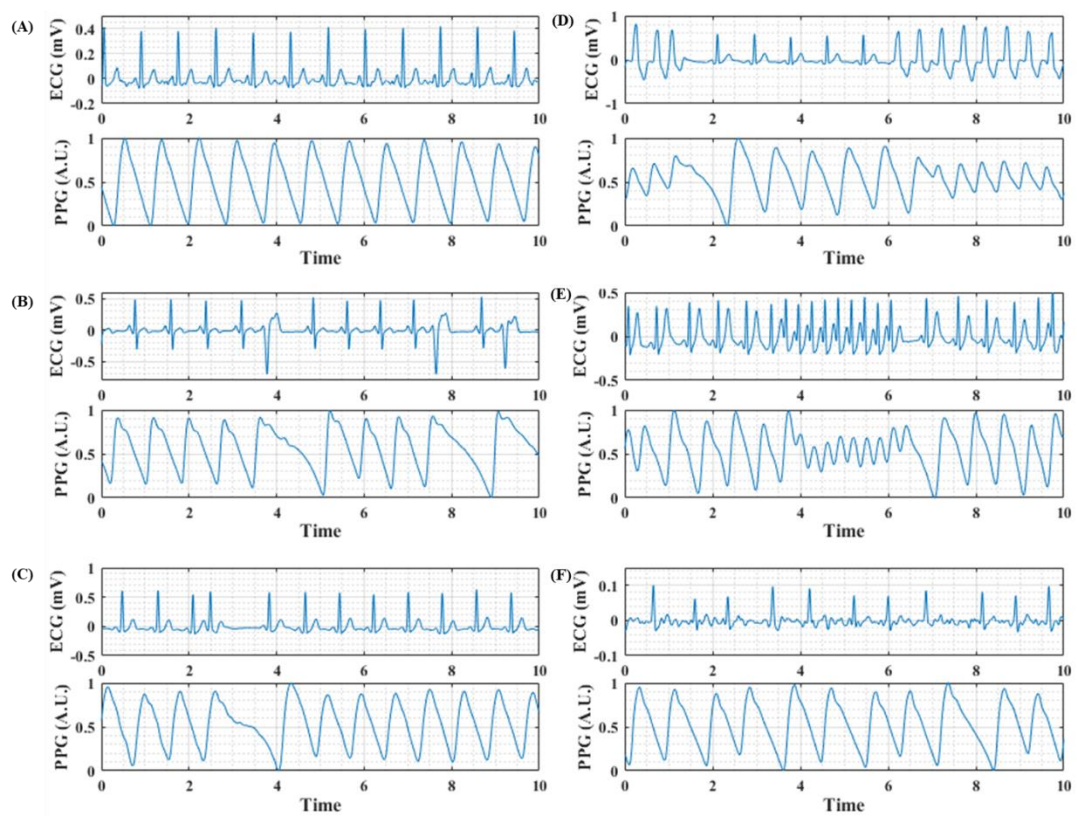
(A) peaks detection and IPI series extraction. (B) PPG waveform features and IPI series features calculation. (C) models development with different machine learning algorithms. PPG, photoplethysmography; IPI, inter-beat intervals; STD, standard deviation; SampEn, sample entropy; ShEn, Shannon entropy; SPI, spectral purity index; CoV, coefficient of variation; nRMSSD, normalized root mean square of successive differences; COSEn, coefficient of sample entropy.

Figure S2. Flowchart of study participants.



ECG and PPG indicate electrocardiogram and photoplethysmography.

Figure S3. Examples of synchronous ECG and PPG signals in sinus rhythm (A), premature ventricular contraction (B), premature atrial contraction (C), ventricular tachycardia (D), supraventricular tachycardia (E), and atrial fibrillation (F).



ECG, electrocardiogram; PPG, photoplethysmography.